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| **Fiscal Year:** | FY 2026 | **Task Last Updated:** | FY 11/08/2025 |
| **PI Name:** | Seidler, Rachael D. Ph.D. | | |
| **Project Title:** | Effect of Head-Down Tilt +/- CO2 on Human Glymphatic Function | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Comments:** | NOTE: PI moved to University of Florida in July 2017; previous affiliation was University of Michigan. | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2020 HERO 80JSC020N0001-FLAGSHIP, OMNIBUS1 Human Research Program: Crew Health Appendix A; Omnibus1-Appendix B |
| **Start Date:** | 01/06/2023 | **End Date:** | 06/30/2026 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Brocato, Becky | **Contact Phone:** | |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 06/30/2026 per V. Lehman/JSC and NSSC information (Ed., 4/30/25)<br>NOTE: End date changed to 06/30/2025 per V. Lehman/JSC (Ed., 3/18/25)<br>NOTE: End date changed to 12/31/2024 per NSSC information (Ed., 7/23/24)<br>NOTE: End date changed to 08/31/2024 per M. Stenger/NASA HHC Element Scientist and C. Ribeiro/NASA HHC (Ed., 1/12/24) | | |
| **Key Personnel Changes/Previous PI:** | N/A | | |
| **COI Name (Institution):** | Richmond, Sutton Ph.D. ( University of Florida, Gainesville )<br>Roy, Arkaprava Ph.D. ( University of Florida, Gainesville ) | | |
| **Grant/Contract No.:** | 80NSSC23K0365 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Spaceflight Associated Neuro-ocular Syndrome (SANS) is a potentially mission-limiting condition for astronauts, yet its underlying pathophysiology is poorly understood. It has been proposed that impairment of the glymphatic system in response to fluid shift or venous congestion could be an underlying cause of SANS. However, there have been no studies to date evaluating human glymphatic function in response to simulated microgravity. While our team is currently evaluating the effect of elevated carbon dioxide (CO2) on human glymphatic function (awards from NASA and the Office of Naval Research), we have yet to evaluate the spaceflight-relevant combination of CO2 and simulated microgravity. (Ed. Note: See related Grant #80NSSC17K0021--PI: Seidler). The latter can be achieved via 6° head-down tilt (HDT), simulating the headward fluid shifts of microgravity. Here, we propose to conduct glymphatic magnetic resonance imaging (MRI) in human subjects over a span of several hours while they are either supine or lying on a 6° foam wedge with head-down tilt. They will also be breathing elevated CO2 to better mimic the conditions on the International Space Station. We will measure the effects of HDT positioning on human glymphatic function (Aim 1) and the combined effects of breathing elevated CO2 and HDT on human glymphatic function (Aim 2). Thus, we will quantify the acute effects HDT with and without elevated CO2. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This project is examining the effects of elevated CO2 combined with and without head down tilt on the brain's waste clearance system (glymphatic system). Understanding the impact that elevated CO2 has on this process may inform environmental health. |
| **Task Progress:** | Spaceflight-Associated Neuro-Ocular Syndrome (SANS) describes a constellation of ocular structural changes exhibited by ~33% of astronauts returning from long-duration spaceflights. These changes include optic disc edema, choroidal folding, globe flattening, and hyperopic refractive error shifts. The etiological mechanisms behind SANS remain insufficiently understood; it is thought to be linked to headward fluid shifts, increased intracranial pressure, altered glymphatic drainage, hypercapnia related volume and pressure disturbances, and a number of other potential mechanisms. Traveling beyond the confines of Earth's atmosphere is accompanied by various health stressors such as microgravity, ionizing radiation, and disrupted circadian rhythms; insight into the various complex interactions of these induced stressors on brain fluid dynamics, glymphatic exchange, and brain health is necessary to develop effective SANS countermeasures.<br>The glymphatic system functions to distribute solutes and clear waste from the brain via cerebrospinal fluid (CSF) circulation and interstitial fluid flow along its perivascular spaces. Glymphatic clearance has been implicated as a key determinant of brain health, acting to remove or redistribute metabolic products, inflammatory and immune-mediated molecules, as well as additional solutes for disposal. Glymphatic dysfunction may cause the failure of interstitial solute clearance or derangement of cranial fluid dynamics resulting from the mismatch between CSF influx and interstitial fluid efflux, with negative impacts on brain homeostasis. It is not known at this point how the environmental stressors of microgravity exposure impact glymphatic function.<br><br>Microgravity exposure produces a fluid shift towards the head, resulting in compression of the top of the brain against the skull, prompting morphological and physiological brain alterations. We have shown that spaceflight results in brain free water redistribution. The resulting changes generate increased intracranial pressures, alter cerebral blood flow, and have been linked to visual impairments. The functional impact of these brain fluid shifts remains to be determined. Head-down tilt (HDT) bedrest has been used as a spaceflight analog; we have reported similar brain positional and fluid shifts in this environment, supporting the notion that these mechanical effects are due to microgravity or rotation of the body in HDT relative to the gravitational vector.<br><br>In addition to altered gravitational effects, astronauts are exposed to increased carbon dioxide (CO2) levels (hypercapnia) in the enclosed environment of the International Space Station. CO2 is a potent vasomodulator; hypercapnia increases both cerebral blood flow and arterial blood pressure. It has been reported that hypercapnia reduces glymphatic exchange in mice. Currently, the effect of elevated CO2 exposure on human glymphatic function remains unknown. Because astronauts are exposed concurrently to microgravity and elevated CO2, here, we propose to study the combinatorial effects of HDT and elevated CO2 on this system.<br><br>In the proposed study, we will investigate the individual and combined effects of simulated microgravity and hypercapnia on human glymphatic function. Using acute HDT bedrest and exposure to elevated CO2, this study will be a critical first step towards understanding the potential interactions of microgravity and elevated CO2 on perivascular glymphatic function and brain health. |
| **Bibliography Type:** | Description: (Last Updated: 03/12/2026) |